Thanh Le-Cong

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Education

2/2023 – Now

The Univerisity of Melbourne
Ph.D Student in Software Engineering

- Advisor: Dr. Bach Le and Prof. Toby Murray
- Funded by the Melbourne Research Scholarship
- Nominated for Google PhD Fellowship 2023

8/2016 – 7/2021

Hanoi University of Science and Technology (HUST)
BE (Honours) in Information Technology

- Advisor: Prof. Huynh Quyet Thang
- Thesis: The Development of Patch Validation for Automated Program Repair.
- Selected for Talented Engineering Program (Top 30 in our cohorts)
- Best Thesis Award AY2021

Research Interests

I am deeply interested in the field of bug/vulnerability detection, management, and repair. My research goal is to develop automated solutions that can leverage recent technological advancements, such as artificial intelligence and automated reasoning, to assist developers in identifying, managing, and fixing bugs and vulnerabilities. Currently, I am working on enhancing the reliablity and trusworthiness of automated bug fixing methods.

Selected Publications

Note:

- *my name with the asterisk, i.e.* ***Thanh Le-Cong**** *denotes that I am the co-first author;*
- *name of my students mentored are underlined.*

⇒ Journal Papers

TSE'23 Truong-Giang Nguyen, **Thanh Le-Cong**, Kang Hong Jin, Ratnadira Widyasari, Chengran Yang, Zhipeng Zhao, Bowen Xu, Jiayuan Zhou, Xin Xia, Ahmed E. Hassan, Bach Le, David Lo; "MiDas: Multi-granularity Model for Vulnerability-fixing Commit Classification"; *IEEE Transaction on Software Engineering (TSE), 2023*

TSE'23 **Thanh Le-Cong**, Bach Le, David Lo, Quyet Thang Huynh, Quang Huy Bui, Nhat Hoa Tran; "Invalidator: Automated Patch Correctness Assessment via Semantic and Syntactic Reasoning"; *IEEE Transaction on Software Engineering (TSE), 2023*

InfoSci'20 Thi Thanh Binh Huynh, Dinh Thanh Pham, Ba Trung Tran, **Thanh Le-Cong**, Minh Hai Phong Le, Ananthram Swami, Thu Lam Bui; "A multifactorial optimization paradigm for linkage tree genetic algorithm"; *Elsevier Information Sciences, 2023*

⇒ Conference Papers

SANER'23 Ratnadira Widyasari, Zhipeng Zhao, **Thanh Le-Cong**, and Hong Jin Kang, David Lo; "Extreme Multi-Label Learning for Topic Recommendation for GitHub Repositories: How far are we?"; *the 30th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), Research Track, 2023 (11 pages, Acceptance Rate = 27%)*

ICSE'23 Yunbo Lyu, **Thanh Le-Cong***, Hong Jin Kang, Ratnadira Widyasari, Zhipeng Zhao, Xuan-Bach D Le, Ming Li, David Lo; "Chronos: Time-Aware Zero-Shot Identification of Libraries from Vulnerability Report"; *the 45th IEEE/ACM International Conference on Software Engineering (ICSE), Technical Track, IEEE, 2023 (11 pages, Acceptance Rate = 26%)*

ESEC/FSE'22 **Thanh Le-Cong**, Kang Hong Jin, Truong-Giang Nguyen, Stefanus Agus Haryono, David Lo, Xuan-Bach D. Le, Quyet-Thang Huynh; "AutoPruner: Transformer-based Call Graph Pruning"; *the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, Research Track, ACM, 2022 (10 pages, Acceptance Rate = 22%)*

ICSME'22 Thanh-Dat Nguyen, **Thanh Le-Cong**, Bach Le, Duc Minh Luong, David Lo, Quyet-Thang Huynh; "FFL: Fine grained Fault Localization for Student Programs via Syntactic and Semantic Reasoning"; *the 38th IEEE International Conference on Software Maintenance and Evolution, Research Track, IEEE, 2022 (10 pages, Acceptance Rate = 23%)*

ISSRE'21 **Thanh Le-Cong**, Bach Le, Quyet Thang Huynh, Phi Le Nguyen; "Usability and Aesthetic: Better Together for Automated Repair of Web Pages"; *the 32nd IEEE International Symposium on Software Reliability Engineering (ISSRE), Research Track. IEEE, 2021 (10 pages, Acceptance Rate = 27.5%)*

⇒ Short/Workshop Papers

ESEC/FSE'22 Truong-Giang Nguyen, **Thanh Le-Cong**, Kang Hong Jin, Xuan-Bach D. Le, David Lo; "VulCurator: A Vulnerability-Fixing Commit Detector"; *the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Demonstration Track, ACM, 2022 (5 pages, Acceptance Rate = 56.5%)*

ICSE'22 Thanh-Dat Nguyen, **Thanh Le-Cong***, ThanhVu H. Nguyen, Bach Le, Huynh Quyet Thang; "Towards the Analysis of Graph Neural Network"; *the 44th IEEE/ACM International Conference on Software Engineering, New Ideas and Emerging Results Track, IEEE, 2022 (5 pages, Acceptance Rate = 26.8%)*

⇒ Works Under Submission

Review Yue Liu, **Thanh Le-Cong**, Ratnadira Widyasari, Chakkrit Tantithamthavorn, Li Li, Bach Le, David Lo; "Refining ChatGPT-Generated Code: Characterizing and Mitigating Code Quality Issues"; *Under Review, ASE 2023*

Review Hung Nguyen, **Thanh Le-Cong**, Hung Le, Loc Nguyen, Bach Le, Quyet-Thang Huynh, David Lo; "Can Code Summarization Succeed with Unseen Projects?"; *Under Review, ASE-NIER 2023*

Review Bowen Xu, Thanh-Dat Nguyen; "Can Code Summarization Succeed with Unseen Projects?"; *Under Review, ASE-NIER 2023*

Review Xin Zhou, Bowen Xu, Kisub Kim, DongGyun Han, **Thanh Le-Cong**, Junda He, Bach Le, and David Lo; "PatchZero: Zero-Shot Automatic Patch Correctness Assessment"; *Under Review, IEEE Transaction on Software Engineering (TSE), 2023*

Experience

2/2023 - Now	**CIS, The University of Melbourne** *Graduate Researcher - Advisor: Dr. Bach Le & Prof. Toby Murray*
1/2022 - 1/2023	**Software Analytics Research Group, Singapore Management University** *Researcher Engineer - Advisor: Prof. David Lo & Dr. Bach Le*
1/2019 - 12/2021	**Intelligent Software Engineering Lab, Hanoi University of Science and Technology** *Researcher Assistant - Advisor: Prof. Huynh Quyet Thang & Dr. Bach Le*
6/2017 - 1/2019	**Modelling, Simulation and Optimization Lab, Hanoi University of Science and Technology** *Researcher Assistant - Advisor: Prof.Huynh Thi Thanh Binh*
7/2019	**Big Data Visualisation Lab, Griffith University** *Researcher Intern - Advisor: Dr. Nguyen Quoc Viet Hung*

Service

⇒ **Program Commitee/Reviewer:** KSE'23
⇒ **Sub-Reviewer:** ICSE'23, SAC'23, KSE'22
⇒ **Student Volunteer:** ICSE'23

Honors and Awards

2023	**Nomination for Google PhD Fellowship (2 nomination per year)** *The University of Melbourne*
2022	**Melbourne Graduate Scholarship** *The University of Melbourne*
2022	**Best Thesis Award (Top 1/600)** *Hanoi University of Science and Technology*
2021	**Best Presentation Award in Thesis Defense Best Thesis Award (Top 30/600)** *Hanoi University of Science and Technology*
2021	**University Scholarship for Academic Excellency (Top 1%)** *Hanoi University of Science and Technology*
2020	**Winner of the Innovative Information Technology Products Contest** *Hanoi University of Science and Technology*
2020	**Runner-up Prize of the Scientific Research Student Competition** *Hanoi University of Science and Technology*
2018	**Winner of the IEEE WCCI Competition on Evolutionary Multi-task Optimization** *IEEE*
2017	**First Prize of the Vietnam Mathematics Olympiad for University Students** *Vietnam Mathematical Society*
2016	**Silver Meldal in the 38th International Mathematic Tournament of Cities in Hanoi** *Vietnam National University*
2016	**Second Prize of the National Mathematical Competition for Excellent High School Students** *Vietnam Ministry of Education and Training*

References

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